CCQM Stage-1 Building Blocks Toolkit

User Manual for Coupling Constants, Decay Constants, and Current-Resolved Form Factors

Stage-1 scope: no decay widths or branching fractions

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This manual explains how to use the Stage-1 CCQM code as a user-facing calculation tool. The user supplies meson parameters, transition parameters, current choices, and a list or grid of q^2 points. The program computes coupling constants g_H, decay constants f_P, f_V, and current-resolved $P \to P$ or $P \to V$ form factors.

Contents

1 What this code does

The program is a Stage-1 building-block calculator for the covariant confined quark model (CCQM). It calculates:

- meson-quark coupling constants g_H, obtained from the compositeness condition $Z_H = 0$,
- decay constants f_P for pseudoscalar mesons and f_V for vector mesons,
- current-resolved transition form factors for pseudoscalar initial mesons:
  - $P_i(0^-) \to P_f(0^-)$,
  - $P_i(0^-) \to V_f(1^-)$.

Important. This package intentionally does *not* compute decay widths, branching ratios, CKM-dependent observables, Wilson-coefficient amplitudes, or angular observables. Those belong to a later Stage-2 observable layer.

2 Files in the package

After unpacking the package, the main directory contains:

File or folder	Purpose
`ccqm.py`	Main user command-line interface. This is the file users run.
`ccqm_stage1_backend.py`	Physics and numerical backend. Advanced users may import this directly.
`requirements.txt`	Python dependencies.
`README.md`	Short quick-start instructions.
`INPUT_GUIDE.md`	Compact input-file guide.
`examples/quick_start.txt`	Minimal $P \to P$ example.
`examples/p_to_v_grid.txt`	Example of $P \to V$ over a q^2 grid.
`examples/full_template.txt`	Larger template showing all input sections.
`sample_output/`	Example output folder from a previous run.

3 Installation

The code is a pure Python command-line tool. It needs Python and the packages listed in `requirements.txt`. From the package folder, run:

```
python -m pip install -r requirements.txt
```

The plotting output uses Matplotlib. The table output uses standard CSV and JSON files.

4 The simplest run

Run the included quick-start example:

```
python ccqm.py run examples/quick_start.txt --output output_quick --precision quick
```

This creates an output folder named `output_quick`. The most useful files to open first are:

```
output_quick/report.txt
output_quick/report.html
output_quick/form_factor_trends.txt
output_quick/form_factor_trends.html
```

5 Useful commands

5.1 Validate an input file

Before running an expensive calculation, check the input file:

```
python ccqm.py validate examples/quick_start.txt
```

The validator catches missing fields, unknown meson names, unsupported currents, and invalid q^2 modes.

5.2 Create a starter template

```
python ccqm.py template my_input.txt
```

This writes a new editable input file.

5.3 Show supported currents

```
python ccqm.py schema
```

This prints the supported current names and the form factors produced by each current.

5.4 Choose precision

```
python ccqm.py run my_input.txt --output results --precision high
```

Precision presets are:

Preset	Quadrature size
`quick`	`n_quad = 4`
`standard`	`n_quad = 8`
`high`	`n_quad = 12`
`very_high`	`n_quad = 16`

For publication-level checks, use `high` or `very_high`. For debugging input files, use `quick`.

6 Input file structure

The input file is a plain text file with sections. It is designed to be readable and editable by non-programmers.

```
[global]
lambda_ir = 0.181
Nc = 3
n_quad = 8

[mesons]
```

```
# name    kind   M       m1      m2      Lambda
B         P      5.279   5.09    0.235   1.88
K         P      0.494   0.424   0.235   1.04

[decay_constants]
B
K

[transition B_to_K]
initial = B
final = K
final_kind = P
m1 = 5.09
m2 = 0.424
m3 = 0.235
currents = scalar, vector, tensor
q2_mode = range
q2_min = 0
q2_max = q2max
q2_points = 10
endpoint = false
```

6.1 The [global] section

Field	Meaning
lambda_ir	Universal infrared cutoff λ, in GeV.
Nc	Number of colors. Usually 3.
n_quad	Gaussian quadrature size. Larger is slower but more accurate.

If a precision preset is passed at the command line, it overrides n_quad.

6.2 The [mesons] section

Each meson line has:

```
name   kind   M   m1   m2   Lambda
```

Column	Meaning
name	Short label used later, such as B, K, Bs, phi.
kind	P for pseudoscalar or V for vector.
M	Meson mass in GeV.
m1,m2	Constituent quark masses in GeV for the meson.
Lambda	Meson size parameter Λ_H in GeV.

The program calculates g_H for every meson in this table.

6.3 The [decay_constants] section

List the mesons for which you want decay constants:

```
[decay_constants]
B
K
```

```
Bs
phi
```

Decay constants do not depend on q^2. They appear in the main report, the decay-constant table, and the decay-constant bar plot.

6.4 Transition sections

A transition block starts with:

```
[transition TRANSITION_NAME]
```

Example:

```
[transition B_to_K]
initial = B
final = K
final_kind = P
m1 = 5.09
m2 = 0.424
m3 = 0.235
currents = scalar, vector, tensor
q2_mode = list
q2_values = 0, 1, 5
```

Field	Meaning
`initial`	Name of the initial pseudoscalar meson from `[mesons]`.
`final`	Name of the final transition meson from `[mesons]`.
`final_kind`	`P` for $P \to P$, or `V` for $P \to V$.
`m1`	Initial active-quark mass.
`m2`	Final active-quark mass.
`m3`	Spectator-quark mass.
`currents`	Comma-separated list of currents to calculate.
`q2_mode`	Either `list` or `range`.

7 Choosing q^2 points

The program supports two q^2 modes.

7.1 Discrete list mode

Use this when you want specific q^2 values:

```
q2_mode = list
q2_values = 0, 1, 5, q2max
```

The keyword `q2max` means:

$$q^2_{\max} = (M_i - M_f)^2.$$

7.2 Grid/range mode

Use this when you want a smooth trend:

```
q2_mode = range
q2_min = 0
q2_max = q2max
q2_points = 20
endpoint = false
```

`endpoint = false` avoids evaluating exactly at the endpoint. This is often safer because some projection bases become numerically singular at special kinematic points.

8 Currents and form factors

The user chooses the currents. The program does not guess the physical decay class. This is intentional: Stage 1 calculates building blocks only.

8.1 Supported currents for $P \to P$

Current name	Operator	Output
`scalar`	1	F_S
`pseudoscalar`	γ_5 or $i\gamma_5$	parity-check amplitude
`vector`	γ^μ	F_+, F_-
`axial`	$\gamma^\mu\gamma_5$	parity-check amplitude
`v_minus_a`	$\gamma^\mu(1-\gamma_5)$	vector form factors plus axial check
`v_plus_a`	$\gamma^\mu(1+\gamma_5)$	vector form factors plus axial check
`tensor`	$i\sigma^{\mu\nu}q_\nu$	F_T

8.2 Supported currents for $P \to V$

Current name	Operator	Output
`scalar`	1	scalar diagnostic amplitude
`pseudoscalar`	γ_5 or $i\gamma_5$	pseudoscalar diagnostic amplitude
`vector`	γ^μ	V
`axial`	$\gamma^\mu\gamma_5$	A_0, A_+, A_-
`v_minus_a`	$\gamma^\mu(1-\gamma_5)$	A_0, A_+, A_-, V, plus BSW-style aliases
`v_plus_a`	$\gamma^\mu(1+\gamma_5)$	current-resolved diagnostic output
`tensor_plus`	$\sigma^{\mu\nu}q_\nu(1+\gamma_5)$	a_0, a_+, g_T
`tensor_minus`	$\sigma^{\mu\nu}q_\nu(1-\gamma_5)$	a_0, a_+, g_T, with convention-dependent signs

Important. For $P \to V$ tensor form factors, the standard tensor basis contains terms proportional to $1/q^2$. Avoid evaluating `tensor_plus` or `tensor_minus` exactly at $q^2 = 0$. Use a small positive q^2 and extrapolate if needed.

9 Output folder

After a run, the output folder has this structure:

```
output_folder/
  results.json
  report.txt
  report.html
  form_factor_trends.txt
  form_factor_trends.html
  tables/
    couplings.csv
    decay_constants.csv
    form_factors.csv
    warnings.csv
    form_factor_trends.csv
    form_factor_trends_wide.csv
  plots/
    decay_constants_bar.png
    ff_*.png
    decay_constants_vs_q2_reference__*.png
```

9.1 Main report

`report.txt` and `report.html` are human-readable summaries. They contain:

- global settings,
- calculated coupling constants,
- decay constants,
- transition summaries,
- warnings.

9.2 Form-factor trend report

`form_factor_trends.txt` and `form_factor_trends.html` are designed for trend checking. They group the output as:

```
Transition: Bs_to_phi
  Current: v_minus_a

    A0
      q2        value
      1.0       ...
      3.0       ...
      5.0       ...

    A_plus
      q2        value
      1.0       ...
      3.0       ...
      5.0       ...
```

This is the easiest report for checking whether a form factor rises, falls, or behaves unexpectedly.

9.3 CSV tables

File	Use
`tables/couplings.csv`	Coupling constants and Z_H checks.
`tables/decay_constants.csv`	Decay constants in GeV and MeV.
`tables/form_factors.csv`	Long-format table: one row per transition/current/-form factor/q^2.
`tables/form_factor_trends.csv`	Trend-friendly long table.
`tables/form_factor_trends_wide.csv`	Wide table: one row per form factor and columns `q2=...`. Best for plotting in Excel, Origin, Python, or gnuplot.
`tables/warnings.csv`	Warnings such as skipped singular tensor projections.

9.4 Plots

The `plots/` folder contains:

- a bar chart for decay constants,
- one form-factor plot per transition/current/form-factor,
- optional horizontal reference plots for decay constants versus q^2.

  Decay constants are not functions of q^2. The reference plots are only visual guides.

10 Example 1: $B \to K$, current-selected $P \to P$

```
[global]
lambda_ir = 0.181
Nc = 3
n_quad = 8

[mesons]
B    P   5.279   5.09    0.235   1.88
K    P   0.494   0.424   0.235   1.04

[decay_constants]
B
K

[transition B_to_K]
initial = B
final = K
final_kind = P
m1 = 5.09
m2 = 0.424
m3 = 0.235
currents = scalar, vector, tensor
q2_mode = range
q2_min = 0
q2_max = q2max
q2_points = 12
endpoint = false
```

Run:

```
python ccqm.py run B_to_K_input.txt --output B_to_K_output --precision high
```

Expected form-factor outputs include F_S, F_+, F_-, and F_T.

11 Example 2: $B_s \to \phi$, current-selected $P \to V$

```
[global]
lambda_ir = 0.181
Nc = 3
n_quad = 8

[mesons]
Bs    P   5.367   5.09    0.424   1.95
phi   V   1.019   0.424   0.424   0.88

[decay_constants]
Bs
phi

[transition Bs_to_phi]
initial = Bs
final = phi
final_kind = V
m1 = 5.09
m2 = 0.424
m3 = 0.424
currents = v_minus_a, tensor_plus
q2_mode = range
q2_min = 1.0
q2_max = q2max
q2_points = 10
endpoint = false
```

Expected outputs include A_0, A_+, A_-, V, and tensor form factors a_0, a_+, g_T.

12 Good user practices

1. Start with `-precision quick` to check that the input file is valid.
2. Then run `-precision standard` or `-precision high` for real values.
3. Use `endpoint = false` for grid calculations unless you specifically need the endpoint.
4. For $P \to V$ tensor currents, avoid exact $q^2 = 0$.
5. Check `tables/warnings.csv` after every run.
6. Use `form_factor_trends.html` or `form_factor_trends_wide.csv` to inspect smooth trends.

13 Troubleshooting

13.1 Unknown meson name

Make sure the name used in a transition block appears exactly in `[mesons]`. Names are case-sensitive.

13.2 Unsupported current

Run:

```
python ccqm.py schema
```

Then copy current names exactly, such as `v_minus_a`, `tensor_plus`, or `scalar`.

13.3 Calculation is slow

Use a smaller precision preset first:

```
python ccqm.py run my_input.txt --output debug_output --precision quick
```

Then increase precision once the input is correct.

13.4 Singular tensor warning

This usually means the requested tensor projection is at $q^2 = 0$ or another problematic kinematic point. Use a small positive value, for example $q^2 = 10^{-4}$, or use a grid with `q2_min = 0.001`.

13.5 Values have tiny imaginary parts

The code converts numerical roundoff such as $0.42 + 10^{-16}i$ into real numbers in reports. If a genuinely complex diagnostic amplitude appears, it is stored in JSON as real and imaginary parts.

14 Recommended project structure for users

For organized research calculations, use one folder per study:

```
my_ccqm_project/
  inputs/
    B_to_K.txt
    Bs_to_phi.txt
  outputs/
    B_to_K/
    Bs_to_phi/
  notes/
    comparison_with_article.md
```

Run examples:

```
python ccqm.py run inputs/B_to_K.txt --output outputs/B_to_K --precision high
python ccqm.py run inputs/Bs_to_phi.txt --output outputs/Bs_to_phi --precision high
```

15 Summary

The Stage-1 CCQM toolkit is meant to be simple for users and systematic for researchers:

- edit a plain text input file,
- choose mesons, currents, and q^2 points,
- run one command,
- read clean reports,

- use CSV tables and plots for checking trends.

It provides the heavy numerical building blocks. The user remains responsible for choosing the physical interpretation of currents and transitions.